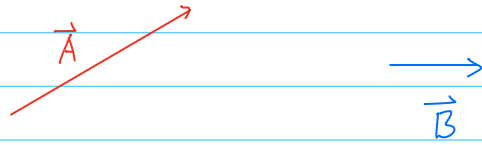
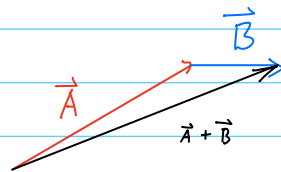


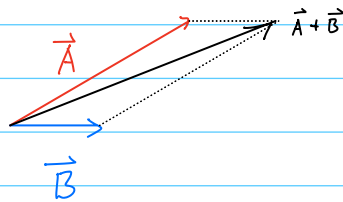
Vector addition & subtraction.



$\vec{A} + \vec{B}$: Method one: "head to tail". join the starting point of B with the end point of A. then connect the starting point of A to the end point of B.

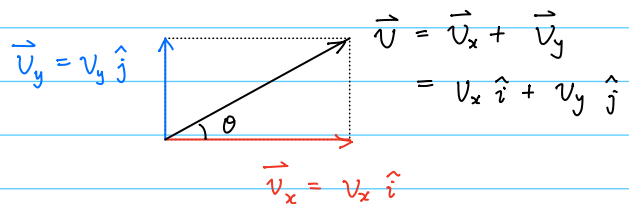


Method two: join the starting points of both, then find the diagonal of the parallelogram bounded by A and B.



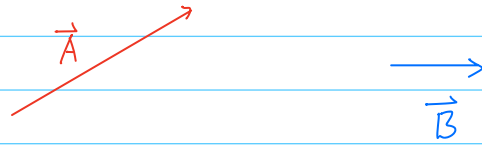
Notes: 1). method one and two are equivalent

2). In special cases where $\vec{A} \perp \vec{B}$, the parallelogram becomes a rectangle. For example:



In this case, because $\vec{v}_x \perp \vec{v}_y$, we can find the magnitude of $\vec{v}$ by $v = \sqrt{v_x^2 + v_y^2}$.

If given θ , we can find $v_x = v \cos \theta$, $v_y = v \sin \theta$, $\tan \theta = \frac{v_y}{v_x}$



$\vec{A} - \vec{B}$: join the starting points of the two, then draw an arrow from the end of the second vector (in this case $\vec{B}$) toward the end of the first vector (in this case $\vec{A}$).

